required hypothesis here and in subsequent uses, including Lemma 2.25 and Corollary 2.30. The contrary permission at the opening of Section 6 is removed.

Lemma 2.24. *Let $\mathfrak{a} \subseteq \mathrm{VF}^n$ be an atomic open polydisc. Then $\mathfrak{a}$ is t-atomic for all $t \in \mathrm{RV}$.*

Proof. For the case $n = 1$, if $\mathfrak{a}$ were not t-atomic then, by Corollary 2.21 and Lemma 2.13, it would contain a point algebraic over the fixed ground base. This point belongs to a finite ground-definable set. Atomicity would then force the whole infinite open disc $\mathfrak{a}$ to lie in that finite set, a contradiction. For the case $n > 1$, by induction on n, $\mathfrak{a}_1$ is t-atomic and for every $a \in \mathfrak{a}_1$, $\mathfrak{a}_{>1}$ is (a, t)-atomic. So $\mathfrak{a}$ cannot have a nonempty proper t-definable subset. □

Partly resolved referee R029 [Gap] – Atomicity proof: algebraic versus definable.
Original source line 886. Lemma 2.13 only gives an algebraic point over the original base, not necessarily a definable point. This suffices: an infinite atomic disc cannot meet a finite base-definable set. Replace 'definable point' at this step by that argument. The final sentence also needs 'nonempty proper t-definable subset'. Finally, indicate why Corollary 2.21 remains applicable whenever previously named disc parameters occur in the base; its stated hypothesis excludes such extra parameters.

Author resolution. The author approves both proof corrections: the point is algebraic over the fixed ground base, so atomicity would place the infinite disc inside a finite ground-definable set; and the final sentence now says "nonempty proper." The statement is unchanged. The separate question about applicability over additional disc-imaginary ground parameters remains open.

Lemma 2.25. *Suppose that $\Gamma(\mathbb{S})$ is nontrivial. Let $A \subseteq \mathrm{VF}^n$ be a set that generates a type and $\mathfrak{b} \subseteq A$ an open (or closed) polydisc. Then $\mathfrak{b}$ is $\ulcorner \mathfrak{b} \urcorner$-atomic.*

Proof. We do induction on n. The base case $n = 1$ is just [18, Lemma 3.4]. For the case $n > 1$, write $\mathfrak{b} = \prod_i \mathfrak{b}_i$, $\mathfrak{b}' = \prod_{i>1} \mathfrak{b}_i$, and $\gamma = \mathrm{rad}(\mathfrak{b}_1)$. For any $a \in \mathfrak{b}_1$, A_a generates a type. So, by Lemma 2.23 and the inductive hypothesis, $\mathfrak{b}'$ is $(a, \gamma, \ulcorner \mathfrak{b}' \urcorner)$-atomic and hence $\ulcorner \mathfrak{b} \urcorner$-atomic. Similarly $\mathfrak{b}_1$ is $\ulcorner \mathfrak{b} \urcorner$-atomic. It follows that $\mathfrak{b}$ must be $\ulcorner \mathfrak{b} \urcorner$-atomic. □

Lemma 2.26. *Let $\mathfrak{a} \subseteq \mathrm{VF}^n$ be an atomic open polydisc and $f : \mathfrak{a} \longrightarrow \mathcal{P}(\mathrm{VF})$ a definable finitary function such that $\mathrm{rv} \restriction f(a)$ is injective for every $a \in \mathfrak{a}$. Then $\mathrm{rv} \circ f$ is constant.*

Proof. This is clear since, by Lemma 2.24, for every $t \in \mathrm{rv}(\bigcup f(\mathfrak{a}))$, $\mathfrak{a}$ is t-atomic and hence f induces a t-definable function $\mathfrak{a} \longrightarrow t^\sharp$. The lemma follows. □

Proposed clarification. A *thin annulus* is a set of the form $\mathfrak{c}(a, \gamma) \setminus \mathfrak{o}(a, \gamma) = \{x \in \mathrm{VF} : \mathrm{val}(x - a) = \gamma\}$, where $a \in \mathrm{VF}$ and $\gamma \in \Gamma$. Equivalently, it is a closed disc with one maximal open subdisc removed.

Lemma 2.27 ([18, Lemma 3.8])**.** *Let $A \subseteq \mathrm{VF}$ be an atomic open disc or an atomic closed disc or an atomic thin annulus. Let $f : A \longrightarrow \mathrm{VF}$ be a definable function. Then $f(A)$ is also atomic of one of these three forms.*

Referee R163 [Clarification; new] – Define thin annuli before using them.
Original source line 907. The term "thin annulus" has not been defined in this manuscript. Specify a closed disc with one maximal open subdisc removed; after choosing the center of the removed subdisc, it is a shell $\{x : \mathrm{val}(x - a) = \gamma\}$. This makes the alternatives here and the later component argument unambiguous.

Lemma 2.28 ([18, Lemma 3.13])**.** *Let $A \subseteq \mathrm{VF}$ be an atomic open disc and $f : A \longrightarrow \mathrm{VF}$ a definable function. Then either f is constant or $f(A)$ is also an atomic open disc.*